US 20250278032A1

(19) United States
(12) Patent Application Publication
KASHIWAYA
(10) Pub. No.: US 2025/0278032 A1
(43) Pub. Date: Sep. 4, 2025

(54) **EUV TRANSMISSIVE MEMBRANE, PROCESSING METHOD THEREOF, AND EXPOSURE METHOD**

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(21) Appl. No.: **19/082,623**

(22) Filed: **Mar. 18, 2025**

Related U.S. Application Data

(63) Continuation of application No. PCT/JP2024/007654, filed on Feb. 29, 2024.

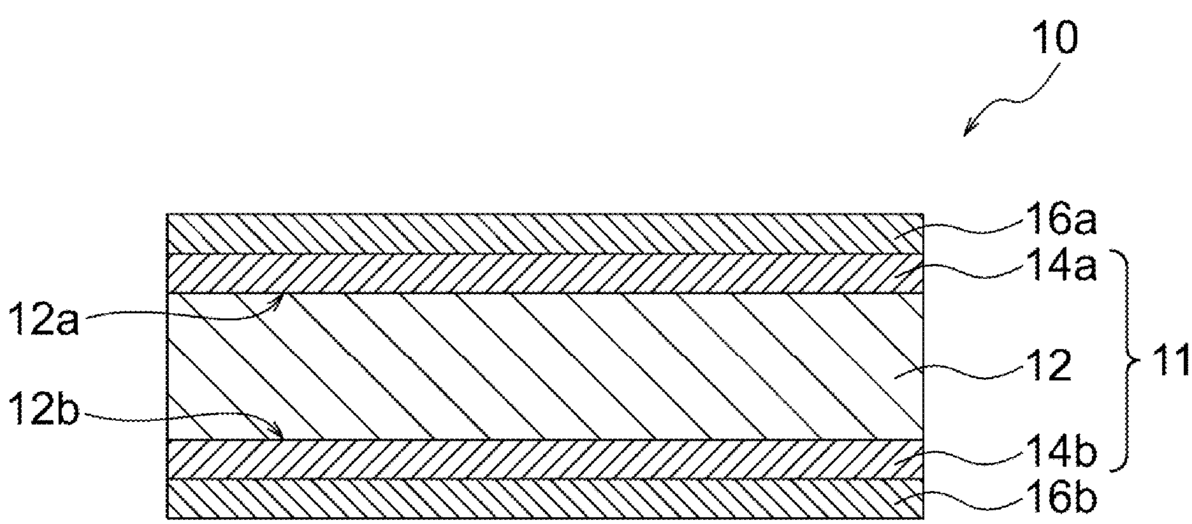
Publication Classification

(51) **Int. Cl.**
G03F 7/00 (2006.01)

(52) **U.S. Cl.**
CPC ***G03F 7/70983*** (2013.01); ***G03F 7/70033*** (2013.01); ***G03F 7/70958*** (2013.01)

(57) **ABSTRACT**

Provided is an EUV transmissive membrane (**10**) having a five-layer configuration consisting of a metallic beryllium layer (**12**) having a first side and a second side; a first nitride layer (**14***a*) including at least one selected from the group consisting of silicon nitride, beryllium nitride, boron nitride, and zirconium nitride, and a first protective layer (**16***a*) containing amorphous carbon, which cover the first side (**12***a*) of the metallic beryllium layer (**12**) in sequence; and a second nitride layer (**14***b*) including at least one selected from the group consisting of silicon nitride, beryllium nitride, boron nitride, and zirconium nitride, and a second protective layer (**16***b*) containing amorphous carbon, which cover the second side (**12***b*) of the metallic beryllium layer (**12**) in sequence. The EUV transmissive membrane has an EUV transmittance of 85% or more at a wavelength of 13.5 nm.



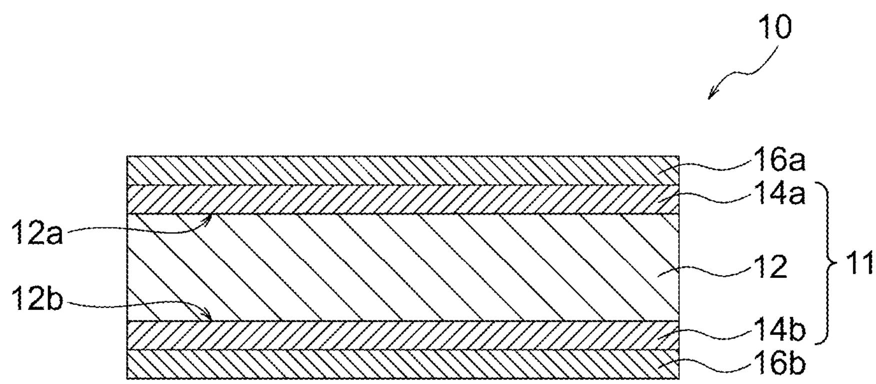


FIG. 1

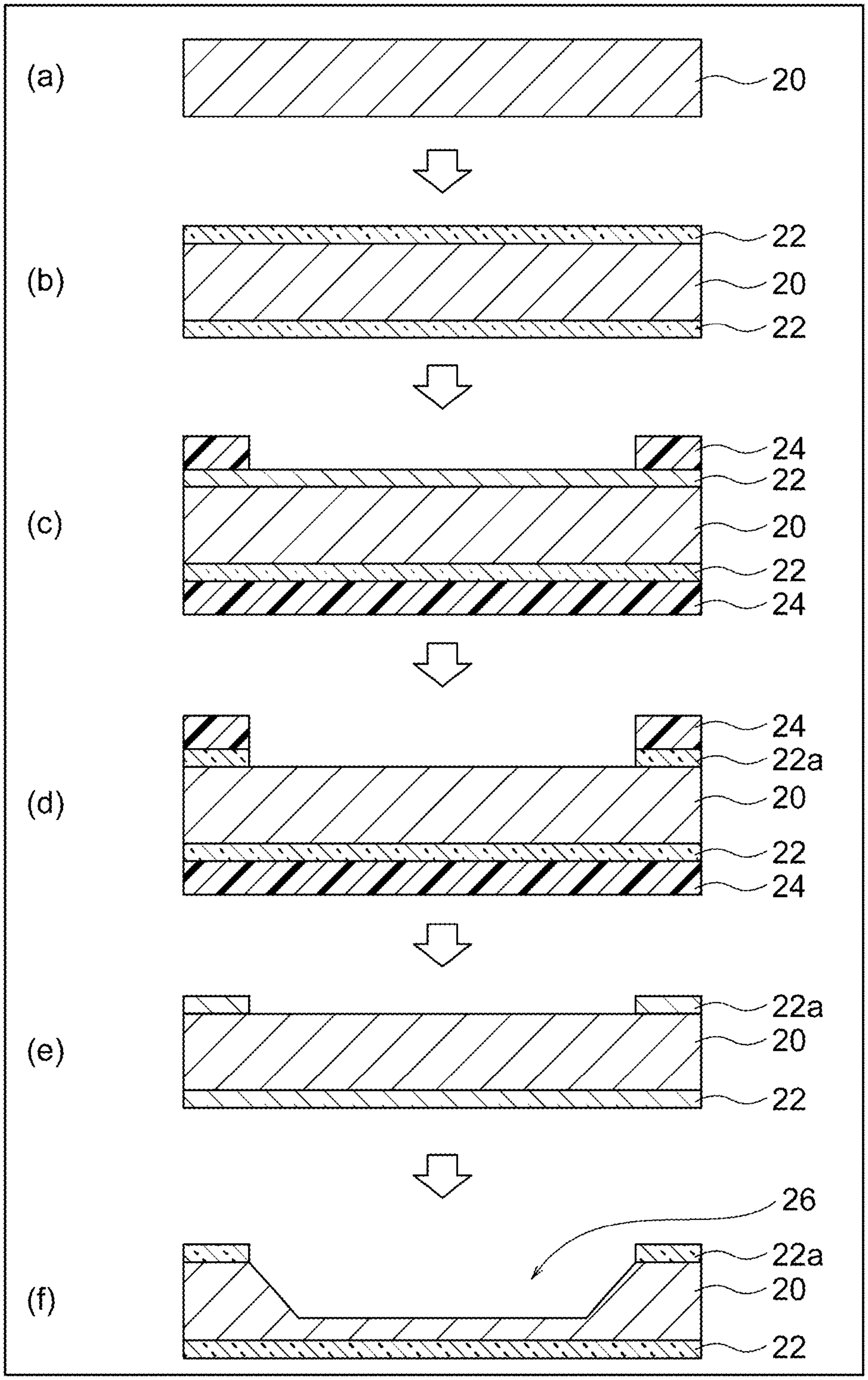

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FIG. 2A

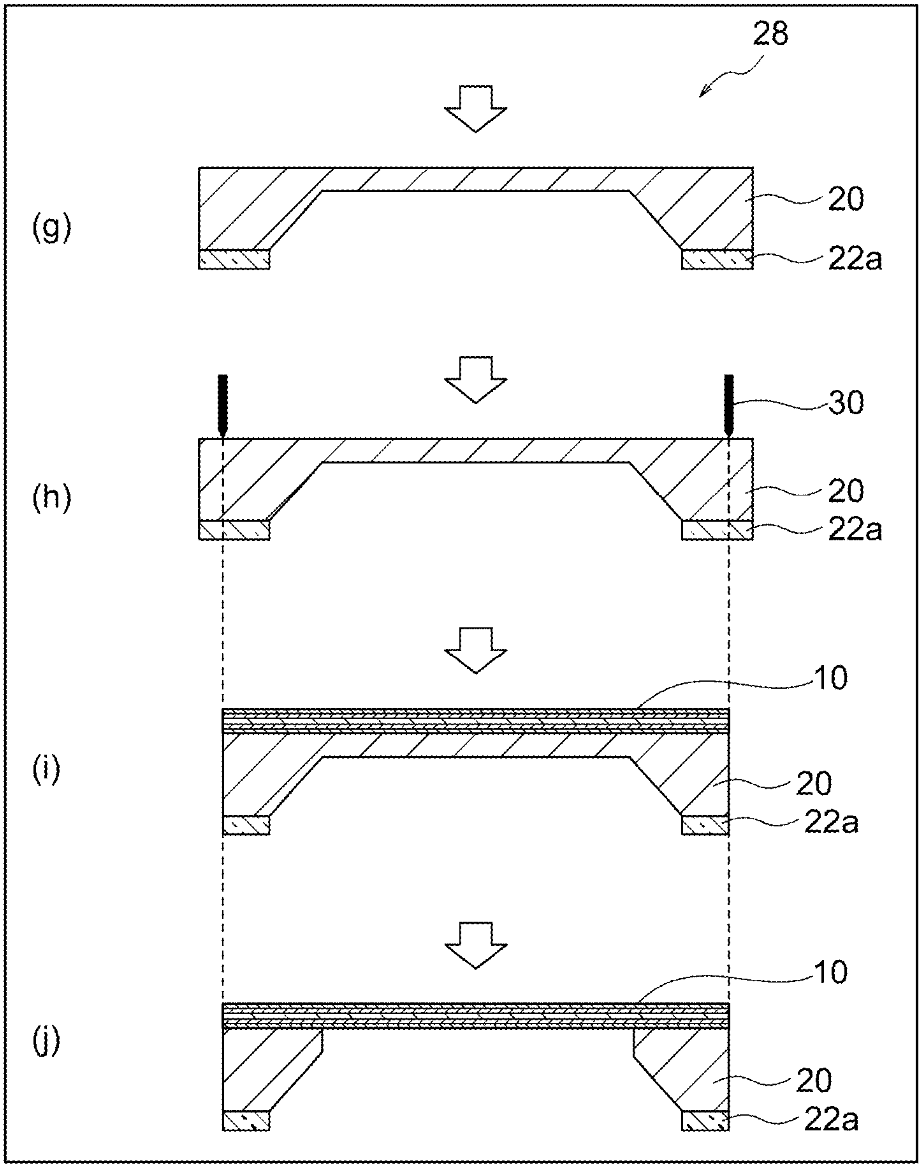

FIG. 2B

EUV TRANSMISSIVE MEMBRANE, PROCESSING METHOD THEREOF, AND EXPOSURE METHOD

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is a continuation application of PCT/JP2024/007654 filed Feb. 29, 2024, the entire contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] The present disclosure relates to an EUV transmissive membrane, processing method thereof, and exposure method.

2. Description of the Related Art

[0003] With miniaturization in semiconductor manufacturing process advancing year by year, various improvements have been made in each step. Particularly, in a photolithography step, extreme ultraviolet (EUV) light having a wavelength of 13.5 nm has begun to be used in place of conventional ArF exposure having a wavelength of 193 nm. As a result, the wavelength was reduced to 1/10 or less at once, and optical properties thereof were completely different. Since there is no material having a high transmittance to EUV light, however, there is still no practical pellicle, which serves as a particle adhesion-preventing membrane of, for example, a photomask (reticle). For this reason, device manufacturers currently cannot use pellicles when manufacturing semiconductor devices.

[0004] Therefore, a pellicle membrane has been developed. It is desirable to use Si, Be, Y, Zr, or the like having a high EUV transmittance as a core material of the pellicle membrane. Patent Literature 1 (JP6858817B) discloses a pellicle membrane including a core layer that contains a material substantially transparent for EUV radiation such as (poly-) Si and a cap layer that contains a material absorbing IR radiation.

[0005] In addition, Patent Literature 2 (JP2020-98227A) discloses a pellicle membrane stretched on one end face of a pellicle frame, the pellicle membrane having a main layer of single-crystal Si, and graphene on one side or both sides of the main layer. It has been proposed that when the main layer has graphene, the pellicle membrane is not damaged during pellicle fabrication and can exhibit sufficient mechanical strength.

CITATION LIST

Patent Literature

Patent Literature 1: JP6858817B

Patent Literature 2: JP2020-98227A

SUMMARY OF THE INVENTION

[0006] The core material having high EUV transmittance used for the pellicle membrane as described above forms a natural oxide membrane of several nm on the surface thereof in air, and the oxide membrane absorbs EUV, resulting in a decrease in EUV transmittance of the pellicle membrane. In particular, Be is a material having high EUV transmittance, but a natural oxide membrane of 2 to 3 nm is considered to be formed on the surface thereof. The transmittance loss due to the oxide membrane is as large as 6 to 9%. In addition, the process of fabricating a pellicle membrane sometimes involves treatment using a gas other than air, such as fluorine or chlorine, an acid, or an alkali solution, which may generate a side reaction membrane on the surface of the core material, thereby decreasing the EUV transmittance. Therefore, it is desirable to form a protective layer on the surface of the core material to suppress the formation of these membranes.

[0007] However, while applying a reaction-inhibiting protective layer on the surface of a core material such as Si and Be can prevent a significant decrease in EUV transmittance of the pellicle membrane, such a protective layer also has lower EUV transmittance than the pure core material, leading to a decrease in the EUV transmittance of the pellicle membrane as a whole. For example, Ru may be used as a protection layer for a Be core material. In the case of a pellicle membrane with a three-layer structure (Ru/Be/Ru) in which 1 to 2 nm of Ru is provided on the front and back surfaces of the Be core material, the Ru protective layer causes an EUV transmission loss of about 3 to 6%, which significantly degrades the performance of the pellicle membrane. Therefore, there is a demand for a protective layer having a small EUV transmission loss. On the other hand, it is desirable that a pellicle membrane has not only high EUV transmittance but also excellent in-plane uniformity of EUV transmittance since better in-plane uniformity of the EUV transmittance results in uniform in-plane exposure and improves homogeneity of a device manufactured through EUV exposure.

[0008] The inventors have recently found that by adopting a five-layer configuration of amorphous carbon layer/nitride layer/metallic beryllium layer/nitride layer/amorphous carbon layer, it is possible to provide an EUV transmissive membrane that exhibits high EUV transmittance and has excellent in-plane uniformity of EUV transmittance while having a protective layer on both sides thereof.

[0009] Accordingly, an object of the present invention is to provide an EUV transmissive membrane that exhibits high EUV transmittance and has excellent in-plane uniformity of EUV transmittance while having a protective layer on both sides thereof. Another object of the present invention is to provide a method of processing an EUV transmissive membrane. Still another object of the present invention is to provide an exposure method using an EUV transmissive membrane.

[0010] The present disclosure provides the following aspects.

Aspect 1

[0011] An EUV transmissive membrane with a five-layer configuration consisting of:

[0012] a metallic beryllium layer having a first side and a second side,

[0013] a first nitride layer that covers the first side of the metallic beryllium layer, wherein the first nitride layer comprises at least one selected from the group consisting of silicon nitride, beryllium nitride, boron nitride, and zirconium nitride,

[0014] a second nitride layer that covers the second side of the metallic beryllium layer, wherein the second nitride layer comprises at least one selected from the group consisting of silicon nitride, beryllium nitride, boron nitride, and zirconium nitride,

[0015] a first protective layer that covers a side of the first nitride layer opposite to the metallic beryllium layer, wherein the first protective layer comprises amorphous carbon, and

[0016] a second protective layer that covers a side of the second nitride layer opposite to the metallic beryllium layer, wherein the second protective layer comprises amorphous carbon,

[0017] wherein the EUV transmissive membrane has an EUV transmittance of 85% or more at a wavelength of 13.5 nm.

Aspect 2

[0018] The EUV transmissive membrane according to aspect 1, wherein the first nitride layer, the metallic beryllium layer, and the second nitride layer constitute a main layer of the EUV transmissive membrane, wherein the main layer has a thickness of 7 to 30 nm.

Aspect 3

[0019] The EUV transmissive membrane according to aspect 1 or 2, wherein the metallic beryllium layer has a thickness of 5 to 25 nm.

Aspect 4

[0020] The EUV transmissive membrane according to any one of aspects 1 to 3, wherein the first nitride layer and the second nitride layer each have a thickness of 1 to 5 nm.

Aspect 5

[0021] The EUV transmissive membrane according to any one of aspects 1 to 4, wherein the first protective layer and the second protective layer each have a thickness of 1 to 10 nm.

Aspect 6

[0022] The EUV transmissive membrane according to aspect 5, wherein the first protective layer and the second protective layer each have a thickness of 2 to 7 nm.

Aspect 7

[0023] The EUV transmissive membrane according to aspect 5, wherein the first protective layer and the second protective layer each have a thickness of 1 nm or more and less than 2 nm.

Aspect 8

[0024] A method of processing an EUV transmissive membrane, comprising the steps of:

[0025] installing the EUV transmissive membrane according to any one of aspects 1 to 7 into an apparatus in which hydrogen plasma and/or hydrogen radicals or oxygen plasma and/or oxygen radicals are generated; and

[0026] bringing hydrogen plasma and/or hydrogen radicals or oxygen plasma and/or oxygen radicals into contact with the EUV transmissive membrane to thin the first protective layer and the second protective layer.

Aspect 9

[0027] An exposure method, comprising the steps of:

[0028] installing the EUV transmissive membrane according to any one of aspects 1 to 7 into an apparatus in which hydrogen plasma and/or hydrogen radicals or oxygen plasma and/or oxygen radicals are generated;

[0029] bringing hydrogen plasma and/or hydrogen radicals or oxygen plasma and/or oxygen radicals into contact with the EUV transmissive membrane to thin the first protective layer and the second protective layer; and

[0030] installing the EUV transmissive membrane in which the first protective layer and the second protective layer are thinned into an EUV exposure apparatus, and then transmitting EUV through the EUV transmissive membrane to perform pattern exposure on a photosensitive substrate plate in the EUV exposure apparatus.

BRIEF DESCRIPTION OF THE DRAWINGS

[0031] FIG. **1** is a schematic cross-sectional view illustrating an embodiment of an EUV transmissive membrane according to the present invention.

[0032] FIG. **2**A is a process flow diagram illustrating the first half of a manufacturing procedure for an EUV transmissive membrane in Examples.

[0033] FIG. **2**B is a process flow diagram illustrating the second half of a manufacturing procedure for an EUV transmissive membrane in Examples.

DETAILED DESCRIPTION OF THE INVENTION

EUV Transmissive Membrane

[0034] FIG. **1** illustrates a schematic cross-sectional view of an EUV transmissive membrane **10** according to an embodiment of the present invention. The EUV transmissive membrane **10** is a membrane with a five-layer configuration consisting of a metallic beryllium layer **12**, a first nitride layer **14***a*, a second nitride layer **14***b*, a first protective layer **16***a*, and a second protective layer **16***b*. The metallic beryllium layer **12** has a first side **12***a* and a second side **12***b*. The first nitride layer **14***a* is a layer that covers the first side **12***a* of the metallic beryllium layer **12** and contains at least one selected from the group consisting of silicon nitride, beryllium nitride, boron nitride, and zirconium nitride. The second nitride layer **14***b* is a layer that covers the second side **12***b* of the metallic beryllium layer **12** and contains at least one selected from the group consisting of silicon nitride, beryllium nitride, boron nitride, and zirconium nitride. The first protective layer **16***a* is a layer that covers a side of the first nitride layer **14***a* opposite to the metallic beryllium layer **12** and contains amorphous carbon. The second protective layer **16***b* is a layer that covers a side of the second nitride layer **14***b* opposite to the metallic beryllium layer **12** and contains amorphous carbon. The EUV transmissive membrane **10** has an EUV transmittance of 85% or more at a wavelength of 13.5 nm. Thus, by adopting a five-layer configuration of first protective layer 16a (amorphous carbon layer)/first nitride layer **14***a*/metallic beryllium layer **12**/second nitride layer **14***b*/second protective layer **16***b* (amorphous carbon layer), it is possible to provide the EUV transmissive membrane **10** that exhibits high EUV transmittance and has excellent in-plane uniformity of EUV transmittance while having the protective layers **16***a* and **16***b* on both sides thereof. As mentioned above, a core material having high EUV transmittance may form a natural oxide membrane of several nm on the surface thereof in air. In addition, the process of fabricating a pellicle membrane sometimes involves treatment using a gas other than air, such as fluorine or chlorine, an acid, or an alkali solution, which may generate a side reaction membrane on the surface of the core material. The formation of such a membrane decreases the EUV transmittance of the pellicle membrane. Therefore, it is desirable to form a protective layer on the surface of the core material to suppress the formation of these membranes. However, applying a reaction-inhibiting protective layer to the surface of the core material leads to a decrease in the EUV transmittance of the pellicle membrane as a whole. On the other hand, it is desirable that a pellicle membrane has not only high EUV transmittance but also excellent in-plane uniformity of EUV transmittance since better in-plane uniformity of the EUV transmittance results in uniform in-plane exposure and improves homogeneity of a device manufactured through EUV exposure. All these problems are successfully solved by the EUV transmissive membrane of the present invention. In other words, the EUV transmissive membrane of the present invention includes a main layer **11** composed of the first nitride layer **14***a*, the metallic beryllium layer **12** and the second nitride layer **14***b*, which has high EUV transmittance, the first protective layer **16***a*, and the second protective layer **16***b*, wherein the protective layers **16***a* and **16***b* cover both sides of the main layer **11**, and these protective layers **16***a* and **16***b* can prevent the formation of a natural oxide membrane or a side reaction membrane which may occur on the surface of the main layer **11** (without protective layers **16***a* and **16***b*) before the pellicle is fabricated, mounted on the EUV exposure apparatus, and subjected to the exposure process. Since the protective layers **16***a* and **16***b* containing amorphous carbon have a small EUV transmission loss, the EUV transmissive membrane **10** of the present invention can exhibit high EUV transmittance even with a five-layer configuration having the protective layers **16***a* and **16***b*. Moreover, the EUV transmissive membrane **10** having the protective layers **16***a* and **16***b* as a five-layer configuration is advantageous in that the in-plane uniformity of EUV transmittance is superior to that of the main layer alone with a three-layer configuration having no protective layers **16***a* and **16***b*. The reason for this is considered that the protective layers **16***a* and **16***b* are formed on both sides, which makes irregularities on both sides of the main layer with a three-layer configuration more even. With this advantage, in the EUV exposure through the EUV transmissive membrane **10**, the in-plane exposure amount becomes uniform, and the homogeneity of the device manufactured through the EUV exposure can be improved.

[0035] As described above, the EUV transmissive membrane **10** has high EUV transmittance. The EUV transmissive membrane **10** has an EUV transmittance of 85% or more, preferably 90% or more, more preferably 91% or more, and even more preferably 92% or more at a wavelength of 13.5 nm. A higher EUV transmittance of the EUV transmissive membrane **10** is desirable, and the upper limit thereof is not particularly limited. Although ideally 100%, the EUV transmissive membrane **10** typically has a EUV transmittance of 95% or less, and more typically 94% or less, for example, 93.5% or less.

[0036] The metallic beryllium layer **12** is a layer containing metallic beryllium as a main component. Here, the "main component" in the metallic beryllium layer **12** means a component that accounts for 50 mol % or more, preferably 70 mol % or more, more preferably 80 mol % or more, and even more preferably 90 mol % or more of the metallic beryllium layer **12**. The metallic beryllium layer **12** may also contain impurities in addition to metal beryllium as a main component. Accordingly, the metallic beryllium layer **12** may be composed of metallic beryllium and inevitable impurities. The thickness of the metallic beryllium layer **12** is preferably 5 to 25 nm, more preferably 7 to 20 nm, and even more preferably 9 to 15 nm.

[0037] The first nitride layer **14***a* and the second nitride layer **14***b* each contain at least one nitride selected from the group consisting of silicon nitride, beryllium nitride, boron nitride, and zirconium nitride. A particularly preferred nitride is silicon nitride. The advantages of providing the first nitride layer **14***a* and the second nitride layer **14***b* on both sides of the metallic beryllium layer **12** are described as follows. For example, when a protective layer of amorphous carbon is provided directly on the main layer, a metallic beryllium layer, the amorphous carbon and metallic beryllium may react to form beryllium carbide. Therefore, the main layer is composed of three layers of first nitride layer **14***a*/metallic beryllium layer **12**/second nitride layer **14***b* and provided with the protective layers **16***a* and **16***b* of amorphous carbon layer on the nitride layers **14***a* and **14***b*. Thus, the reaction between amorphous carbon and metallic beryllium can be prevented, that is, the reaction between the main layer and the protective layer can be suppressed. The terms "silicon nitride", "beryllium nitride", "boron nitride", and "zirconium nitride" as used herein each mean a comprehensive composition that allows not only a stoichiometric composition such as Si_3N_4, Be_3N_2, BN, and ZrN, but also a non-stoichiometric composition such as Si_3N_{4-x}, wherein $0<x<4$, Be_3N_{2-x}, wherein $0<x<2$, BN_x, wherein $0<x<1$, and ZrN_x, wherein $0<x<1$. The first nitride layer **14***a* and the second nitride layer **14***b* each preferably have a thickness of 1 to 5 nm, and more preferably 1 to 3 nm.

[0038] The first nitride layer **14***a*, the metallic beryllium layer **12**, and the second nitride layer **14***b* constitute a main layer **11** of the EUV transmissive membrane. The main layer **11** contributes to the realization of high EUV transmittance while ensuring basic functions as a pellicle membrane (such as a particle adhesion preventing function). The thickness of the main layer **11** is preferably 7 to 30 nm, more preferably 9 to 26 nm, and even more preferably 11 to 21 nm.

[0039] When the first nitride layer **14***a* and/or the second nitride layer **14***b* each contain beryllium nitride, the main layer **11** preferably has a nitrogen concentration gradient region where nitrogen concentration decreases as closer to the metallic beryllium layer **12**. In other words, when the composition of beryllium nitride may include from the stoichiometric composition such as Be_3N_2 to the non-stoichiometric composition such as Be_3N_{2-x}, wherein $0<x<2$, as described above, the beryllium nitride constituting the beryllium nitride layer preferably has a gradient composition that is richer in beryllium as closer to the metallic beryllium layer

12. As a result, it is possible to improve the adhesion between the nitride layers **14**_a_ and **14**_b_ (i.e., beryllium nitride layer) constituting the main layer **11** and the metallic beryllium layer **12** and relieve the generation of stress caused by the difference in thermal expansion between these layers. That is, it is possible to improve the adhesion between these layers to suppress delamination, and to make delamination difficult as a thermal expansion relaxation layer between these layers in the case of absorbing EUV light and becoming a high temperature. The thickness of the nitrogen concentration gradient region is preferably smaller than that of each of the nitride layers **14**_a_ and **14**_b_. In other words, the entire thickness of each of the nitride layers **14**_a_ and **14**_b_ does not need to be in the nitrogen concentration gradient region. For example, it is preferable that only a part of the thickness of each of the nitride layers **14**_a_ and **14**_b_, for example, a region of preferably 10 to 70% and more preferably 15 to 50% of the thickness of each of the nitride layers **14**_a_ and **14**_b_ is the nitrogen concentration gradient region.

[0040] Each of the first protective layer **16**_a_ and the second protective layer **16**_b_ is a layer containing amorphous carbon. The layer containing amorphous carbon is advantageous in that, in addition to exhibiting high protection performance against various agents (e.g., fluorine-based etching agent with strong reactivity) used in a pellicle membrane fabricating process (e.g., free-standing membrane forming step), the layer has high EUV transmittance and reduces the impact of the residue on the main layer **11** when the protective layers **16**_a_ and **16**_b_ are thinned or removed, and the protective layers **16**_a_ and **16**_b_ are easily thinned or removed. The first protective layer **16**_a_ and the second protective layer **16**_b_ preferably contains amorphous carbon as a main component, and more preferably is composed of amorphous carbon. In general, amorphous carbon does not have a completely disordered atomic arrangement but microscopically has a crystal structure (i.e., it has crystallites). Such crystallites are often arranged in a disordered manner, making them amorphous as a whole. In particular, diamond-like carbon (DLC) refers to a material containing many crystallites with a cubic four-coordinated structure, such as diamond, while graphite-like carbon (GLC) refers to a material containing many crystallites with a planar three-coordinated structure, such as graphite. The first protective layer **16**_a_ and the second protective layer **16**_b_ may be carbon with a completely disordered atomic arrangement, carbon having crystallites in a disordered manner, or even amorphous carbon that is not completely dense but rather contains fine pores. Here, the "main component" in the first protective layer **16**_a_ and the second protective layer **16**_b_ means a component that accounts for 50% by weight or more, preferably 60% by weight or more, more preferably 70% by weight or more, and even more preferably 80% by weight or more of the total weight of the first protective layer **16**_a_ and the second protective layer **16**_b_. Meanwhile, the first protective layer **16**_a_ and the second protective layer **16**_b_ may be composed only of amorphous carbon.

[0041] The first protective layer **16**_a_ and the second protective layer **16**_b_ each preferably have a thickness of 1 to 10 nm, more preferably 2 to 7 nm, and even more preferably 3 to 5 nm. In such a range, high EUV transmittance and improvement of in-plane uniformity of EUV transmittance can be more effectively realized. In addition, it is desirable that the protective layers **16**_a_ and **16**_b_ are thick in order to protect the EUV transmissive membrane **10** (especially, main layer **11**) from various agents (e.g., fluorine-based etching agent with strong reactivity) used in the pellicle membrane fabricating process (e.g., free-standing membrane forming step). However, the EUV transmittance of the EUV transmissive membrane **10** decreases when the thickness of the membrane is increased. In this regard, when the thickness of the membrane is within the above range, it is possible to realize a good balance between high protection performance against various agents and high EUV transmittance. Meanwhile, in order to realize even higher EUV transmittance while ensuring the minimum protection performance, it is also effective to extremely reduce the thickness of each of the first protective layer **16**_a_ and the second protective layer **16**_b_ to, for example, preferably 1 nm or more and less than 2 nm, more preferably 1 nm or more and less than 1.5 nm.

[0042] In the EUV transmissive membrane **10**, the main region for transmitting EUV is preferably in a form of the free-standing membrane. In other words, it is preferable that the substrate (e.g., Si substrate) used at the time of deposition remains as a border only at the outer edge of the EUV transmissive membrane **10**, that is, no substrate (e.g., Si substrate) remains in the main region other than the outer edge. In short, the main region preferably consists of the main layer **11** and the protective layer **16**_a_, **16**_b_.

Manufacturing Method

[0043] After a laminated membrane to be an EUV transmissive membrane is formed on a Si substrate, the EUV transmissive membrane according to the present invention can be fabricated by removing an unnecessary portion of the Si substrate through etching to form a free-standing membrane. Accordingly, the main portion of the EUV transmissive membrane is in the form of the free-standing membrane in which no Si substrate remains as described above.

(1) Preparation of Si Substrate

[0044] First, a Si substrate for forming a laminated membrane thereon is prepared. After the laminated membrane composed of the second protective layer **16**_b_, the second nitride layer **14**_b_, the metallic beryllium layer **12**, the first nitride layer **14**_a_, and the first protective layer **16**_a_ is formed on the Si substrate, the main region (i.e., a region to be a free-standing membrane) other than the outer edge of the Si substrate is removed by etching. Accordingly, it is desirable to reduce the thickness of the Si substrate in the region to be formed into the free-standing membrane in advance in order to perform the etching efficiently in a short time. Therefore, it is desirable that a mask corresponding to the EUV transmission shape is formed on the Si substrate by employing a normal semiconductor process, and the Si substrate is etched by wet etching to reduce the thickness of the main region of the Si substrate to a predetermined thickness. The wet-etched Si substrate is cleaned and dried to prepare a Si substrate having a cavity formed by wet etching. The wet etching mask may be made of any material that is corrosion resistance to Si wet etchant, for example, SiO_2 is suitable for use. In addition, the wet etchant is not particularly limited as long as it is capable of etching Si. For example, TMAH (tetramethylammonium hydroxide) is preferred because it can be used under appropriate conditions, and very good anisotropic etching can be performed on Si.

(2) Formation of Laminated Membrane

[0045] The laminated membrane composed of the second protective layer **16***b*, the second nitride layer **14***b*, the metallic beryllium layer **12**, the first nitride layer **14***a*, and the first protective layer **16***a* is formed in order on the Si substrate. The laminated membrane may be formed by any deposition method. An example of the preferred deposition method is the sputtering method. It is preferable that the metallic beryllium layer **12** is fabricated by sputtering using a pure Be target. It is preferable that the amorphous carbon membrane as the first protective layer **16***a* and the second protective layer **16***b* is done by sputtering using a graphite target.

[0046] The first nitride layer **14***a* and the second nitride layer **14***b* are also preferably fabricated by sputtering. For example, the nitride layers **14***a* and **14***b* may be formed by (i) forming a Si, Be, B or Zr membrane by sputtering using a Si, Be, B, or Zr target and then irradiating the membrane with nitrogen plasma to cause a nitriding reaction of Si, Be, B, or Zr. The first nitride layer **14***a* and the second nitride layer **14***b* may also be fabricated by (ii) reactive sputtering. The reactive sputtering can be performed, for example, by adding nitrogen gas to the chamber during sputtering using a Si, Be, B, or Zr target, whereby Si, Be, B, or Zr and nitrogen react to each other to generate silicon nitride, beryllium nitride, boron nitride, or zirconium nitride. Alternatively, the nitride layers **14***a* and **14***b* may be directly formed by (iii) sputtering using a Si_3N_4, Be_3N_2, BN, or ZrN target. Nitrogen gas may be added to the chamber during the sputtering, whereby Si, Be, B, or Zr and nitrogen react to each other as in the case of the reactive sputtering (ii) above to promote the generation of silicon nitride, beryllium nitride, boron nitride, or zirconium nitride.

[0047] The method of forming the metallic beryllium layer **12**, the nitride layers **14***a* and **14***b*, and the protective layers **16***a* and **16***b* is not limited thereto. The metallic beryllium layer **12**, the nitride layers **14***a* and **14***b*, and the protective layers **16***a* and **16***b* may be formed in a one-chamber sputtering apparatus as in Examples described later, or a multi-chamber sputtering apparatus may be used to form the metallic beryllium layer **12**, the nitride layers **14***a* and **14***b*, and the protective layers **16***a* and **16***b* in separate chambers.

[0048] In the case where the first nitride layer **14***a* and the second nitride layer **14***b* are beryllium nitride layers including a nitrogen concentration gradient region, that is, in the case of forming a nitrogen concentration gradient region in the main layer **11** with a three-layer structure of beryllium nitride/beryllium/beryllium nitride, when depositing the beryllium nitride membrane and metallic beryllium, the introduction of nitrogen gas may be stopped and switched to metallic beryllium deposition while continuing sputtering using a pure Be target with nitrogen gas in the chamber. In this way, a region is formed in which the nitrogen concentration in the film-deposited membrane decreases in the thickness direction as the concentration of nitrogen gas in the chamber decreases. On the other hand, in the case of switching the metallic beryllium to the beryllium nitride, the nitrogen concentration gradient region can be formed by starting the introduction of nitrogen gas in the middle of the process while sputtering is performed, contrary to the above. The thickness of the nitrogen concentration gradient region can be controlled by adjusting the time for which the nitrogen gas concentration is changed.

(3) Free-standing Membrane Formation

[0049] An unnecessary portion of the Si substrate other than the outer edge of the Si substrate where the composite membrane is formed, which is left as a border, is removed by etching to make the composite membrane free-standing. Etching of Si may be performed by any method, but etching with XeF_2 is preferred.

Processing Method

[0050] Optionally, the manufactured EUV transmissive membrane **10** may be processed to be thin. As described above, by making the first protective layer **16***a* and the second protective layer **16***b* thin, it is possible to realize even higher EUV transmittance while ensuring the minimum protection performance. Particularly, after the free-standing membrane forming step using a fluorine-based etching agent such as XeF_2 described above, the first protective layer **16***a* and the second protective layer **16***b* no longer require a strong protective function against the fluorine-based etching agent, and it is sufficient if the first protective layer **16***a* and the second protective layer **16***b* have a mild protective function such as oxidation resistance. Therefore, after the EUV transmissive membrane **10** is made into a free-standing membrane, it may be advantageous to further enhance the EUV transmittance by making the protective layers **16***a* and **16***b* thin. The processing for making the EUV transmissive membrane **10** thin is preferably performed by installing the EUV transmissive membrane **10** into an apparatus in which hydrogen plasma and/or hydrogen radicals or oxygen plasma and/or oxygen radicals are generated, then bringing hydrogen plasma and/or hydrogen radicals or oxygen plasma and/or oxygen radicals into contact with the EUV transmissive membrane **10**. The amorphous carbon membrane, which serves as the first protective layer **16***a* and the second protective layer **16***b*, is exposed to a hydrogen plasma and/or hydrogen radical atmosphere or an oxygen plasma and/or oxygen radical atmosphere, whereby C on the surface of the amorphous carbon membrane reacts with H or O to partly remove the amorphous carbon membrane. As a result, the first protective layer **16***a* and the second protective layer **16***b* can be made thin.

Exposure Method

[0051] As described above, by making the first protective layer **16***a* and the second protective layer **16***b* thin, it is possible to realize even higher EUV transmittance while ensuring the minimum protection performance. Therefore, it is preferable to make the first protective layer **16***a* and the second protective layer **16***b* thin before exposure. From this viewpoint, a preferred exposure method includes the steps of: installing the EUV transmissive membrane **10** into an apparatus in which hydrogen plasma and/or hydrogen radicals or oxygen plasma and/or oxygen radicals are generated; bringing hydrogen plasma and/or hydrogen radicals or oxygen plasma and/or oxygen radicals into contact with the EUV transmissive membrane **10** to thin the first protective layer **16***a* and the second protective layer **16***b*; and installing the EUV transmissive membrane **10** in which the first protective layer **16***a* and the second protective layer **16***b* are thinned in an EUV exposure apparatus, and then transmitting EUV through the EUV transmissive membrane **10** to perform pattern exposure on a photosensitive substrate plate in the EUV exposure apparatus.

EXAMPLES

[0052] The present invention will be described in more detail with reference to the following examples. However, the present invention is not limited to the following examples.

Example 1

[0053] According to the procedures illustrated in FIGS. 2A and 2B, a composite free-standing membrane (EUV transmissive membrane) with a five-layer configuration of amorphous carbon/silicon nitride/Be/silicon nitride/amorphous carbon was fabricated as follows.

(1) Preparation of Si Substrate

[0054] A Si wafer **20** having a diameter of 8 inches (20.32 cm) was prepared (FIG. **2**A (a)). A SiO_2 membrane **22** having a thickness of 50 nm was formed on both sides of the Si wafer **20** by thermal oxidation (FIG. **2**A (b)). Resist was applied to both sides of the Si wafer **20**, and a resist mask **24** for SiO_2 etching was formed by exposure and development so that a 110 mm×145 mm resist hole was created on one side (FIG. **2**A (c)). An exposed portion of the SiO_2 membrane **22** was etched and removed by wet-etching one side of the substrate with hydrofluoric acid to fabricate a SiO_2 mask **22***a* (FIG. **2**A (d)). The resist mask **24** for SiO_2 etching was removed with an ashing apparatus (FIG. **2**A (e)). Si was then wet-etched with a TMAH solution. This etching was performed only for an etching time to obtain a target Si substrate having thickness of 50 μm with an etching rate measured in advance (FIG. **2**A (f)). Finally, the SiO_2 membrane **22** formed on the surface not subjected to Si etching was removed and cleaned with hydrofluoric acid to prepare a Si substrate **28** (FIG. **2**B (g)). The Si substrate outline may be diced with a laser **30**, if necessary (FIG. **2**B (h)), to achieve the desired shape (FIG. **2**B (i)). In this way, a 110 mm×145 mm cavity **26** was provided at the center of the 8-inch (20.32 cm) Si wafer **20** to prepare the Si substrate **28** having a Si thickness of 50 μm in the cavity **26** portion.

(2) Formation of Composite Membrane

[0055] On the Si substrate **28** including the cavity **26** obtained in (1) above, a composite membrane with a five-layer structure of amorphous carbon/silicon nitride/Be/silicon nitride/amorphous carbon was formed as follows (FIG. **2**B (i)). First, the Si substrate plate **28** was set in a multi-target sputtering apparatus, and a graphite target, Si_3N_4 target and a pure Be target were attached thereto. A chamber was evacuated, the graphite target was used to carry out sputtering only with argon gas at an internal pressure of 0.3 Pa, and the sputtering was terminated at the time when 2 nm of diamond-like carbon (DLC) was layer-deposited as amorphous carbon. Subsequently, the chamber was evacuated again, the Si_3N_4 target was used to carry out sputtering with a gas obtained by adding **20**% of nitrogen gas to argon gas at an internal pressure of 0.3 Pa, and the sputtering was terminated at the time when 2 nm of Si_3N_4 was layer-deposited. Furthermore, the chamber was evacuated again, the pure Be target was used to carry out sputtering only with argon gas at an internal pressure of 0.5 Pa, and the sputtering was terminated at the time when 20 nm of beryllium was layer-deposited. Subsequently, the chamber was evacuated again, the Si_3N_4 target was used to carry out sputtering with a gas obtained by adding 20% of nitrogen gas to argon gas at an internal pressure of 0.3 Pa, and the sputtering was terminated at the time when 2 nm of Si_3N_4 was layer-deposited. Thereafter, sputtering was performed using the graphite target in the same manner as in the first step, and the sputtering was terminated at the time when 2 nm of amorphous carbon was layer-deposited. In this manner, a composite membrane with 2 nm of amorphous carbon (C)/2 nm of silicon nitride (Si_3N_{4-x})/**20** nm of beryllium (Be)/2 nm of silicon nitride (Si_3N_{4-x})/2 nm of amorphous carbon (C) was formed as the EUV transmissive membrane **10**. In other words, the EUV transmissive membrane **10** has a five-layer configuration consisting of a main layer composed of three layers of a first silicon nitride layer, a metallic beryllium layer, and a second silicon nitride layer, a first protective layer, and a second protective layer, wherein the first protective layer and the second protective layer containing amorphous carbon as a main component are formed on both sides of the main layer.

(3) Free-standing Membrane Formation

[0056] The Si substrate **28** with the EUV transmissive membrane **10** prepared in (2) above was set in a chamber of an XeF_2 etcher capable of processing an 8-inch (20.32 cm) substrate. The chamber was sufficiently evacuated. At this time, if moisture remains in the chamber, the moisture reacts with the XeF_2 gas to generate hydrofluoric acid, and corrosion of the etcher or unexpected etching occurs. Therefore, the sufficient evacuation was performed. If necessary, vacuuming and nitrogen gas introduction were repeated in the chamber to reduce residual moisture in the chamber. When the sufficient evacuation was achieved, a valve between a XeF_2 material tank and a preliminary space was opened. As a result, XeF_2 was sublimated, and XeF_2 gas was also accumulated in the preliminary space. When the XeF_2 gas was sufficiently accumulated in the preliminary space, the valve between the preliminary space and the chamber was opened to introduce the XeF_2 gas into the chamber. The XeF_2 gas was decomposed into Xe and F, and F reacted with Si to generate SiF_4. Since the boiling point of SiF_4 was −95° C., SiF_4 generated was rapidly evaporated, causing a reaction of F with the newly exposed Si substrate. When the Si etching proceeded and F in the chamber decreased, the chamber was evacuated, and the XeF_2 gas was introduced into the chamber again to perform the etching. In this manner, the evacuation, the introduction of the XeF_2 gas, and the etching were repeated, and the etching was continued until the Si substrate **28** corresponding to the portion to be formed into the free-standing membrane disappeared. When the Si substrate of the unnecessary portion disappeared, the etching was terminated. In this way, a composite free-standing membrane with a five-layer configuration having a border **20** made of Si is obtained as the EUV transmissive membrane **10** (FIG. **2**B (j)).

Example 2

[0057] An EUV transmissive membrane **10** having a border **20** made of Si was fabricated in the same manner as in Example 1. Thereafter, the amorphous carbon membrane exposed on both sides of the EUV transmissive membrane **10** was etched with hydrogen plasma to reduce the thickness of each amorphous carbon layer to 1 nm. In this manner, a composite free-standing membrane with a five-layer configuration consisting of 1 nm of amorphous carbon (C)/2 nm of silicon nitride (Si_3N_{4-x})/20 nm of beryllium (Be)/2 nm of silicon nitride (Si_3N_{4-x})/1 nm of amorphous carbon (C), which had a border **20** made of Si, was obtained as the EUV transmissive membrane **10**.

Example 3 (Comparison)

[0058] An EUV transmissive membrane **10** having a border **20** made of Si was fabricated in the same manner as in Example 1. Thereafter, the amorphous carbon membrane exposed on both sides of the EUV transmissive membrane **10** was etched with hydrogen plasma to remove each amorphous carbon layer. In this manner, a composite free-standing membrane with a three-layer configuration consisting of 2 nm of silicon nitride (Si_3N_{4-x})/20 nm of beryllium (Be)/2 nm of silicon nitride (Si_3N_{4-x}), which had a border **20** made of Si, was obtained as the EUV transmissive membrane **10**.

EUV Transmittance and In-plane Uniformity thereof

[0059] EUV light was irradiated onto a free-standing membrane as the EUV transmissive membrane fabricated in Examples 1 to 3 at a power of 600 W in a hydrogen atmosphere of 20 Pa for 15 minutes, and then the amount of transmitted EUV light was measured with a sensor. When the EUV transmittance was determined by comparing the obtained measurement value with a value obtained by directly measuring the amount of EUV light with a sensor without using the EUV transmissive membrane, the results shown in Table 1 were obtained. At this time, a spot of the EUV light used for measuring the transmittance had an elliptical shape of 0.5 mm×0.2 mm, and the EUV transmittance was measured within the spot size. Therefore, in order to evaluate the in-plane uniformity of the EUV transmittance, the EUV transmittance at each place was determined by measuring the EUV transmittance while moving the spot of the EUV light, and the value of the in-plane variation was calculated from the data as a value three times the standard deviation. Accordingly, the smaller value of the in-plane variation refers to better in-plane uniformity of the EUV transmittance.

TABLE 1

	Component and thickness (nm) of each layer					EUV properties	
	First protective layer C	First nitride layer Si_3N_{4-x}	Metallic beryllium layer Be	Second nitride layer Si_3N_{4-x}	Second protective layer C	EUV transmittance (%)	In-plane variation in EUV transmittance (%)
Example 1	2	2	20	2	2	91.1	0.2
Example 2	1	2	20	2	1	92.6	0.3
Example 3*	None	2	20	2	None	93.8	0.5

*represents a comparative example.

[0060] As can be seen from the results, when the amorphous carbon membrane is left as in Examples 1 and 2, the EUV transmittance is slightly lowered, while the in-plane variation in the EUV transmittance is small, that is, the in-plane uniformity becomes excellent. On the other hand, when the amorphous carbon membrane is etched and removed as in Example 3, the transmittance increases, while the in-plane variation in the EUV transmittance is large, that is, the in-plane uniformity is deteriorated. In that sense, it can be said that by making the amorphous carbon membrane thin and leaving it as in Example 2, the in-plane uniformity of the EUV transmittance can be improved more than in Example 3 while mostly maintaining a high level of EUV transmittance close to that in Example 3. Although a higher EUV transmittance results in shorter exposure time, that is, higher throughput can be advantageously achieved, a smaller in-plane variation in the EUV transmittance results in uniform in-plane exposure amount and improves homogeneity of the device. In other words, the present invention can provide an EUV transmissive membrane suitable for improving homogeneity of the device while realizing high EUV transmittance.

What is claimed is:

1. An EUV transmissive membrane with a five-layer configuration consisting of:

- a metallic beryllium layer having a first side and a second side,
- a first nitride layer that covers the first side of the metallic beryllium layer, wherein the first nitride layer comprises at least one selected from the group consisting of silicon nitride, beryllium nitride, boron nitride, and zirconium nitride,
- a second nitride layer that covers the second side of the metallic beryllium layer, wherein the second nitride layer comprises at least one selected from the group consisting of silicon nitride, beryllium nitride, boron nitride, and zirconium nitride,
- a first protective layer that covers a side of the first nitride layer opposite to the metallic beryllium layer, wherein the first protective layer comprises amorphous carbon, and
- a second protective layer that covers a side of the second nitride layer opposite to the metallic beryllium layer, wherein the second protective layer comprises amorphous carbon,

wherein the EUV transmissive membrane has an EUV transmittance of 85% or more at a wavelength of 13.5 nm.

2. The EUV transmissive membrane according to claim **1**, wherein the first nitride layer, the metallic beryllium layer, and the second nitride layer constitute a main layer of the EUV transmissive membrane, wherein the main layer has a thickness of 7 to 30 nm.

3. The EUV transmissive membrane according to claim **1**, wherein the metallic beryllium layer has a thickness of 5 to 25 nm.

4. The EUV transmissive membrane according to claim **1**, wherein the first nitride layer and the second nitride layer each have a thickness of 1 to 5 nm.

5. The EUV transmissive membrane according to claim **1**, wherein the first protective layer and the second protective layer each have a thickness of 1 to 10 nm.

6. The EUV transmissive membrane according to claim **5**, wherein the first protective layer and the second protective layer each have a thickness of 2 to 7 nm.

7. The EUV transmissive membrane according to claim **5**, wherein the first protective layer and the second protective layer each have a thickness of 1 nm or more and less than 2 nm.

8. A method of processing an EUV transmissive membrane, comprising the steps of:

installing the EUV transmissive membrane according to claim **1** into an apparatus in which hydrogen plasma and/or hydrogen radicals or oxygen plasma and/or oxygen radicals are generated; and

bringing hydrogen plasma and/or hydrogen radicals or oxygen plasma and/or oxygen radicals into contact with the EUV transmissive membrane to thin the first protective layer and the second protective layer.

9. An exposure method, comprising the steps of:

installing the EUV transmissive membrane according to claim **1** into an apparatus in which hydrogen plasma and/or hydrogen radicals or oxygen plasma and/or oxygen radicals are generated;

bringing hydrogen plasma and/or hydrogen radicals or oxygen plasma and/or oxygen radicals into contact with the EUV transmissive membrane to thin the first protective layer and the second protective layer; and

installing the EUV transmissive membrane in which the first protective layer and the second protective layer are thinned into an EUV exposure apparatus, and then transmitting EUV through the EUV transmissive membrane to perform pattern exposure on a photosensitive substrate plate in the EUV exposure apparatus.

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